Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Qin, Yi-Xian Ph.D.		
Project Title:	A Non-Invasive Scanning Confocal Ultrasonic Diagnostic System for Bone Quality		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:	National Institutes of Health/National Institute on Aging	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Gruber, Barry (SUNY- The State University of New York) Rubin, Clinton (Research Foundation of SUNY)		
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	The bone loss which parallels extended space missions represent serious threat to astronaut health, both during flight and on return to gravitational fields. Early diagnosis of osteopenia and its associated bone quality would enable prompt treatment and thus dramatically reduce the risk of fracture. The goal of this project is to develop a new technology for monitoring bone quality of humans during long-term space missions and on Earth, for assessing both bone density and strength. This will lead to a better understand of the progressive adaptation of bone loss in astronauts subject to microgravity and aging populations, and the ensuing musculoskeletal complications such as osteoporosis. It has been shown that bone loss induced in the long duration microgravity environment proceeds at an average rate of 2% per month, ranging highest loss in the skeleton of the lower body. While these results are detected only when astronauts returned to Earth, the rate of bone loss during space mission is remained unclear. Recent advances in quantitative		

Task Description:	ultrasound have enabled a true potential assessment of bone quality, including not only bone density but also the mechanical strength. In this study, funded by the NSBRI, we are able to develop a scanning confocal acoustic diagnostic (SCAD) system capable of generating acoustic images at the regions of interest (e.g., in the human calcaneus). This portable SCAD system is capable of generating non-invasive, high-resolution ultrasound (US) attenuation and velocity maps of bone, and thus determining the relationship between ultrasonic specific parameters and bone mineral density (BMD), and bone strength and bone's physical properties (i.e., stiffness and modulus). Both animal and human trials indicate strong correlations between SCAD and micro-CT determined parameters of bone's material properties, including BMD (R=0.87) and yield strength (R=0.9). The developed technology has demonstrated its uniqueness in image base, confocal scanning, capable to selecting region of interests, and assessing both quantity (i.e., density) and quality (i.e., stiffness) of bone. This system is relevant not only for ground-based determination of bone's physical properties, but can effectively be used in the space environment for determining even subtle changes in density and strength during extended flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Closely related to microgravity induced bone loss, osteoporosis is a reduction in bone mass or density that leads to deteriorated and fragile bones. It is the leading cause of bone fractures in postmenopausal women and in the elderly population for both men and women. About 13 to 18 percent of women aged 50 years and older and 3 to 6 percent of men aged 50 years and older have osteoporosis in the US alone. These rates correspond to 4~6 million women, and 1~2 million men who suffer from osteoporosis. One-third of women over 65 will have vertebral fractures and 90% of women aged 75 and older have radiographic evidence of osteoporosis. Thus, approximately a total of 24 million people suffer from osteoporosis in the United States, with an estimated annual direct cost of over $16 billion to national health programs. Hence, an early diagnosis that can predict fracture risk and result in prompt treatment is extremely important. At present, osteoporotic bone loss is commonly assessed by measures of bone quantity using bone mineral density (BMD) measurement that reflects in vivo conditions of bone density and mass. Noninvasive measurements of bone mass would be of value in predicting the risk of fracture, in assessing the severity of the disease, and in following response to treatment. Several methods are available for the measurement of bone mass, with the most commonly used methods being DEXA and computed tomography (CT). While DEXA is currently a gold standard technique used because of its relative precision (~2%), and whole body and/or multi-site imaging ability (spine, hip, wrist and total skeleton), a certain percent of bone mass must be lost before significant radiation attenuation occurs. If not only bone mineral density but bone quality, e.g. stiffness and/or modulus, can be monitored or determined instantly during the space mission, then one can better understand the skeleton adaptation on a daily basis. In the case of osteopenia and/or osteoporosis, fractures can occur without a singular traumatic event. Recently, new methods in quantitative ultrasound (QUS) have emerged with the potential to estimate cancellous bone modulus more directly. The primary advantage of ultrasonic techniques is that it is capable of measuring not only bone quantity, but also bone quality, i.e., estimation of the mechanical property of bone. Over the past 15 years, a number of research approaches have been developed to quantitate bone mass and structural stiffness using quantitative ultrasound. QUS is simple, inexpensive, non-invasive and free of ionizing radiation. Importantly, the measurements obtained do not provide information about the integrity of the trabecular architecture, or information about the mechanical properties of the composing trabeculae. As a result, an improved diagnostic tool is needed to evaluate both the quantity and the quality of bone, which will help in the early detection and therefore the possible prevention and treatment of this disease. The data have demonstrated that SCAD system has potentials to impact the diagnosis of bone quality on this matter.
Task Progress:	The objectives of this project are focused on developing a new scanning ultrasound system to provide an insight in non-invasively diagnostic of osteoporosis and predict bone loss in the matter of both density and strength (bone quality). The research team focused on technology development of the Scanning Confocal Acoustic Diagnostic (SCAD) system (S.A.1 & 3) and validation between SCAD determined acoustic parameters and bone quality data (S.A.2). Human cadaver and in vivo subject testing were also performed (S.A.4). Technology Development: A system design, including hardware and software, of an diagnostic prototype was established which includes acoustic, electrical, control and mechanical components. This micro-controller guided acoustic scanning technology greatly reduces the scan time in several minutes. Ultrasonic attenuation and velocity images are obtained and calculated, e.g., gray scale or virtual color mapping. SCAD as noninvasive modality for animal bone quality assessment: The ability of SCAD in non-invasively evaluating bone quality and quantity was tested in a large group of ex vivo sheep bone samples. The results of ultrasonic attenuation and velocity were correlated with mechanical moduli of the sample. Using SCAN system, micro-CT and mechanical testing, strong correlations are observed between SCAD determined BUA and micro-CT determined parameters. SCAD used for human calcaneus bone quality assessment: The feasibility of SCAD assessment for bone quality in the real body region, which includes soft tissue, cortical bone and different surface morphology, is evaluated in cadaver calcaneus. BUA and ultrasound velocity determined from region of interest (ROI) have been performed. Bone samples were further tested for structural and strength parameters using micro-CT and mechanical testing in the extracted cylindrical samples from ROI. Strong correlations were found between ultrasound parameters and bone volume fraction (BV/TV)(R2 > 0.64) using combined parameters of BUA and UV in linear regression which ultrasound images determined parameters predict structural, e.g., structure morphological index (SMI) (R2 = 0.86), and strength modulus (R2 = 0.64). These results suggest that high-resolution acoustic mapping is capable of predicting bone quantity and quality non-invasively. Structural property parameters of trabeculi, e.g., BMD and BV/TV, are better represented by BUA, while ultrasonic wave velocity has a strong agreement with bone's strength property, e.g., modulus. Ultrasonic imaging has shown the great potential to be used as in vivo diagnostic modality for assessing skeletal disorder, i.e., osteoporosis.
Bibliography Type:	Description: (Last Updated: 02/17/2021)
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